

Designing for a Unified Experience: A New Perspective and a Case Study

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ABSTRACT

Our chapter begins with an analysis of the challenges faced by today's enterprise software environment and their impacts on user experience (UX). Today's enterprise software environment in large companies is a hybrid environment, where vendor solutions and home grown applications co-exist, in combination of with legacy and new technologies, plus possible siloed solutions. Such a hybrid and siloed environment poses a number of serious challenges to delivering an acceptable experience to the enterprise users. The challenges would eventually have a negative impact on user productivity and business objectives if left unmet.

A review of the existing literature on how to improve UX in such an environment shows that there is no effective approach proposed or practiced to address the challenges. Based on the review and discussions, we propose a comprehensive definition of unified experience with five attributes to address the challenges: consistency, efficiency, personalization, integration, and optimization. Furthermore, we propose an implementation framework for achieving unified experience.

To demonstrate the advantages of the proposed approach, we present a case study that focuses on the unified experience design of a suite of software applications in the enterprise supply chain domain. The case study describes how we executed the approach and demonstrates the advantage by implementing the unified experience approach.

At the close of the chapter, we conclude that the proposed approach can help address the challenges that we currently face. Finally, we discuss the lessons learned and how they may be applied to developing future strategies for implementing unified experience.

Key words: user experience, enterprise software, unified experience,

The goal of ergonomics practices in designing software applications is to deliver optimal user experience (UX) to end users (Dekker and Nyce 2004; Dul et al. 2012; Karwowski 2005). In this chapter, we first discuss the challenges faced by today's enterprise software environment and their impacts on UX. After that, we review the existing literature on how to improve UX in such an environment. Based on the review and following discussions, we propose a comprehensive definition of unified experience and an implementation framework to address the challenges. In order to illustrate the benefits of such an approach, we then present a case study. Finally, we discuss the lessons learned and how they may be applied to developing future strategies for implementing unified experience.

1. CHALLENGES IN ENTERPRISE SOFTWARE AND THEIR IMPACTS ON UX

1.1 Challenges

Many large companies have grown either organically or through acquisition over the last few decades. In many cases, they have ended up in a situation where they have to maintain a number of platforms as well as hundreds of applications across product portfolios for use by internal employees. These applications include CRM (customer relation management), SCM (supply chain management), finance systems, human resources management, BI (business intelligence), production support systems, utility and productivity applications, and etc. Employees rely on these applications to do their daily jobs.

These huge suites are the result of mergers, acquisitions, home-grown applications developed for specific internal requirements, and vendor solutions purchased over the years. Kloczek (2012) described the situation very well: "Unless you're developing completely new products at a startup, you likely work in an organization that has accumulated years of legacy design and development in its products." During the growth of these enterprise solutions over years, these applications were delivered to employees with a variety of business, legal, and user needs, as well as divergent technological requirements. Sometimes, technical patchworks have forced business teams to create workarounds in their processes (Baturan, 2015). Lack of cross-

department visibility may have resulted in duplication, redundancy, and inefficiency or lack of integration across applications in terms of data, business processes, user interfaces, and end user support.

Concerns about business efficiency and cost reduction will see ever more enterprise applications taking advantage of technological advances such as cloud-based SaaS solutions. This will mean more third party applications from a variety of vendors are entering the enterprise software landscape.

As a result, we see a hybrid environment of enterprise software where in-house legacy applications and vendor solutions (e.g. SaaS) co-exist. Furthermore, these applications may be combined with a number of silos in terms of technical platforms and business processes/workflows. These applications are often incompatible, with new patches having simply been laid over the old ones. Specific business requirements and legal constraints have sometimes even resulted in cases of contradictory business logic and interconnected use.

1.2 Impacts on UX

From UX perspective, enterprise users rely on a set of applications, rather than a single one, to accomplish their daily tasks through pre-defined business processes and rules. They interact with these applications through a variety of user touch points, including user interface (UI), business process or work flow, application capability, setup (device) procedure, application performance, data (BI), marketing, online help, user support, integration across applications, and access across channels (mobile, desktop, etc.). UX in today's digital landscape is distributed across an array of such touch points. UX is not simply influenced by a single application or an individual touch point, but instead by how all these applications and touch points are combined to provide a seamless and continued experience to end users. If any of these touch points were to break down, it would create negative experience (Xu, 2014A & 2016).

When working within such a hybrid environment of enterprise applications—in many cases siloed solutions, as described above—enterprise users will often struggle when learning a suite of related applications because of differences in how they look, function, operate, and integrate with each other (Abel, 2015). This sometimes leads to an overall broken user experience (Klocek,

2012). As a result, user support call volume may increase, or employee productivity, business decision-making efficiency, or employee satisfaction may decrease. Eventually, companies' business objectives might be negatively impacted. Sometimes an organization may have UX functions, but it is often the case that the maximum UX professionals can manage, within the limited time allotted, is merely to ensure that the visual design is consistent across applications. Even when it comes to visual design, many vendor applications do not allow customization or do not have a UI that enables the end user to configure the visuals for branding purposes.

Generally speaking, there are several organizational factors that affect the situation (Rogowski, 2012). First, whether or not upper management teams have the mindset or necessary experience to assess UX in the first place. Employee feedback needs to be received early in the decision-making process so that decisions won't be left to technical teams. Secondly, whether or not a company has adequate standards. The standards should not only cover basic features like visual styles and the layouts of applications, but also include well-defined standards for more complex UX practices such as vendor solutions and business process integration across domains and applications. Lastly, even when companies have well-defined standards, cross-organization/-division governance is also important since designers are able to follow them, allowing them to coordinate effectively with other teams.

Thus, the hybrid environment of today's enterprise software poses a number of serious challenges to delivering an acceptable experience to the enterprise users, challenges that would eventually have a negative impact on their productivity and business objectives if left unmet. As user interactions span an increasingly complex array of fractured user touch points across various vendor and home-grown solutions, platforms, and devices, companies need to take action to improve the experience of users across all touch points.

2. SEEKING SOLUTIONS

2.1 Previous Initial Efforts

Companies are waking up to the realization that UX is one of the main drivers of competitive advantage today (Rogowski, 2012; Al-Riyami, 2014). Given the high stakes, a number of groups

have attempted to address the challenges posed by UX in enterprise software. The easiest thing to deal with, and hence the first thing people have tried to do, is delivering a visually uniform experience for the user across UIs. For instance, Google has adopted a “material design style” across applications (Google, 2015). This approach primarily focuses on the visual interaction design of applications. Cooper’s approach also places more emphasis on interaction design across applications in order to foster consistent interactions and positively transform experience across applications (Goodwin, 2009).

Efforts that manage to ensure that the look (visual style) and feel (interactions) of applications are consistent are considered to deliver uniform experience across applications. But this kind of UX approach has been criticized for only dealing with the problem in a superficial manner (Rogowski, 2012). This is because the goal is only to deliver the same visual design and verbatim messaging across applications and platforms (e.g. desktop, mobile) and to make the same depth and breadth of content with all its functionality available in each. It only delivers a one-size-fits-all experience that basically ignores the platform and device capabilities and the specific needs in each end user context.

Rogowski (2012) proposed a unified customer experience approach in the digital marketing domain. He argues that unified customer experiences don’t have to be uniform. Customers need experiences that are the right size for the touchpoint and its context. Specifically, he recommends that the goal of a company’s cross-customer channel efforts be to create a seamless network of interaction points that enables end users to choose where they want to interact and provides consistency of information across touchpoints as well as continuity when transitioning between them, all while maintaining a coherent brand personality. Although this approach is limited to the domain of digital marketing, the approach does address UX beyond the UI portion of applications and goes deeper into the experience by providing targeted and personalized content appropriate to the context.

O’Toole (2013) approaches the challenges in a yet deeper way by unifying the architecture. He believes an integrated system includes multiple components that must be carefully fitted together to achieve the required functionality. This implies running many different servers and databases alongside some middleware that ties the whole operation together. A good enterprise app suite should take advantage of an integrative approach by designing an enterprise cloud and a unified

in-memory architecture specifically to allow applications and data to exist together as a unified whole. He further believes that a unified architecture is important because it is the “engine house” that the end users care about. Although O'Toole intends to address the challenges at the level of the backend database and architecture, he gives no details to demonstrate how the unified architecture actually works in tandem with other applications—such as the front end side—to achieve a better experience.

Finstad, Xu, et al. (2009) attempted to bridge the gap between enterprise software and end users. Specifically, they approached the challenges by calling for a partnership between the vendors of enterprise software and its users—besides necessary in-house UX work. By working together, suppliers can deliver an experience that better meets the needs of enterprises and their users.

In order to deliver an optimized experience across various user touch points for enterprise applications, Xu (2013, 2014A) proposed a user experience design (UXD) framework that was intended to address the gaps in current user-centered design practices that primarily focus on UI design and its usability, and the lack of a systematic approach towards addressing total user experience across all user touch points. UI is only one of the user touch points encountered when interacting with applications. The UXD framework approaches the challenges faced in the enterprise software from a broader UX ecosystem context. Such an approach provides a broader perspective from which we can tackle the challenges.

2.2 A Promising Concept– Unified Experience

As just discussed, initial efforts have attempted to tackle the challenges from different angles. Kloeck's model appears to be the most promising one so far, although it is still incomplete (Kloeck, 2012). As illustrated in Fig. 1, Kloeck proposes a hierarchy of efforts as a strategy for fixing the broken experience in the enterprise software environment, which starts with surface improvements, goes progressively deeper into structural issues, and ends with a big organizational shift. Specifically, he recommends the need for a series of efforts to fixing the broken experiences we face today in enterprise software, although he does not provide a clear definition of what “unified experience” actually is. In principle, Kloeck's approach echoes what Forrester called “unified customer experience” in the digital marketing domain (Rogowski, 2012). Forrester believes that a unified experience in the content management domain should

ensure that content, functionality, and a coherent brand personality matches user expectations, tasks, and contexts across touch points.

In Klocek's model (see Fig. 1), the least amount of effort that can be made is shown at the bottom of the pyramid: one can simply take a consistent visual design approach to help bring different products under a shared brand experience. The next step is to make the interaction behavior consistent. This requires disciplined reuse of interaction patterns through widgets, as well as unifying the nomenclature and conceptual frameworks. The third step is to rework the products even further. That means evaluating the current products against the user's needs and goals as well as looking for ways to eliminate work and to simplify the patterns. Moving up to the fourth level, designing a *unified experience* requires looking at the big picture, reevaluating the internal product silos in the organization, and reconsidering the ideal workflow for individuals and between job roles. This kind of work takes deep organizational commitment and a strong mandate. The last step is to transform the organization itself. Enterprise applications are conceived and developed in a particular organizational culture.

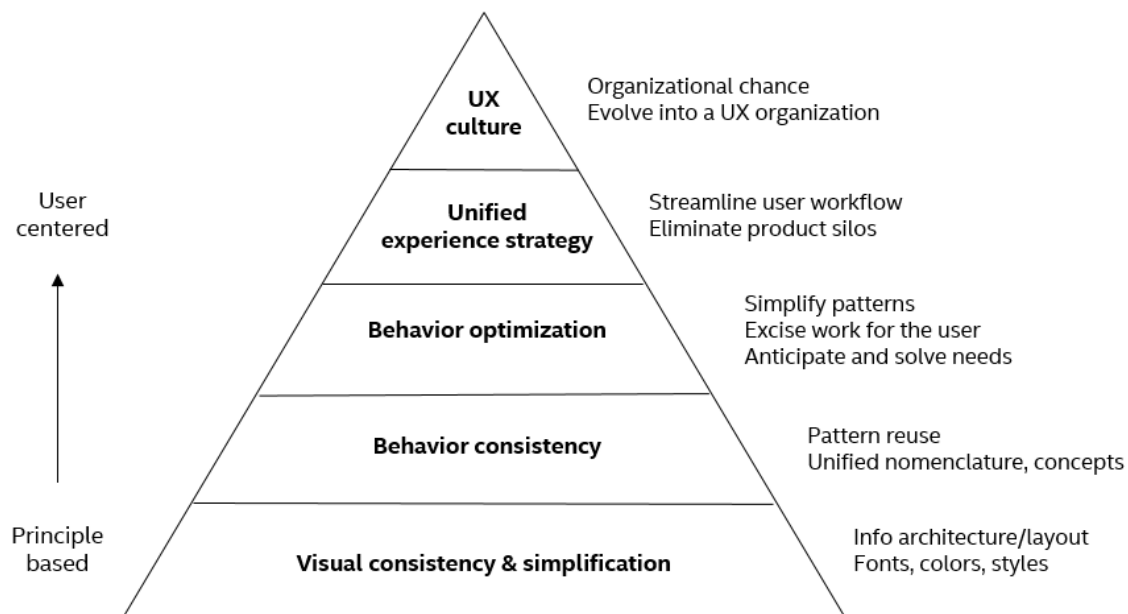


Figure 1. The hierarchy of effort to fix a broken user experience (Klocek, 2012).

Klocek's approach and the unified experience concept are encouraging. However, there are a number of weaknesses in Klocek's model. First of all, Klocek over emphasizes the importance of visual design and considers it as the foundation of the whole efforts. We believe that the benefits

of visual consistency and simplification are limited if one is aiming to achieve a unified experience. Visual consistency can only achieve surface effects. For instance, if the business processes and work flows behind the applications are not streamlined across applications, whatever you do to the UI, the gain in UX is going to be limited.

Secondly, while the model does provide a strategy to address the challenges, it does not specifically fit the hybrid environments of enterprise software that we intend to address here. Specifically, the efforts at levels 2 and 3 assume companies could modify and customize vendor products by implementing specific interaction patterns across applications for consistent and optimized behaviors. As Klocek acknowledges: “Hopefully, any individual product will have internally consistent patterns.” But this is not realistic. In most cases, customizing vendor products, even just for the UI layer, is either technically unfeasible, or very costly for companies during future upgrades.

Third, business process redesign is placed too late in the model because the hierarchical model is supposed to define a sequence of actions to be taken during implementation. However, no matter what they do, companies will not be able to deliver optimal experience if the fundamental business processes and workflows are not streamlined and optimized. In addition, integration across applications /platforms, which is necessary in order to achieve seamless experience across applications and platforms, is not considered in the model. Optimization across user touch points is not considered in the model either.

Finally, in practice, the process of achieving a unified experience may not progress in a linear fashion as outlined in the model—and that goes for either a bottom-up or a top-down approach. The top-down approach requires fundamental culture change and organizational commitment, which is unlikely to happen at the beginning. The bottom-up approach, which starts with consistent visual design across applications, won’t bring enough benefits.

In summary, although people have devoted a lot of effort to addressing the problems across different “layers” in order to create a better experience, we believe that none of them are comprehensive enough to effectively address the UX challenges we are facing in the enterprise software domain. We believe that a more strategic, comprehensive, and flexible approach should be designed.

3. A UNIFIED EXPERIENCE FRAMEWORK

3.1 A Comprehensive Definition of a Unified Experience

Based on the weaknesses identified in the previous approaches discussed above, we first propose a comprehensive concept for unified experience in Fig. 2. The intention is to address the challenges and gaps that comprise the broken experiences found in today's enterprise software environment.

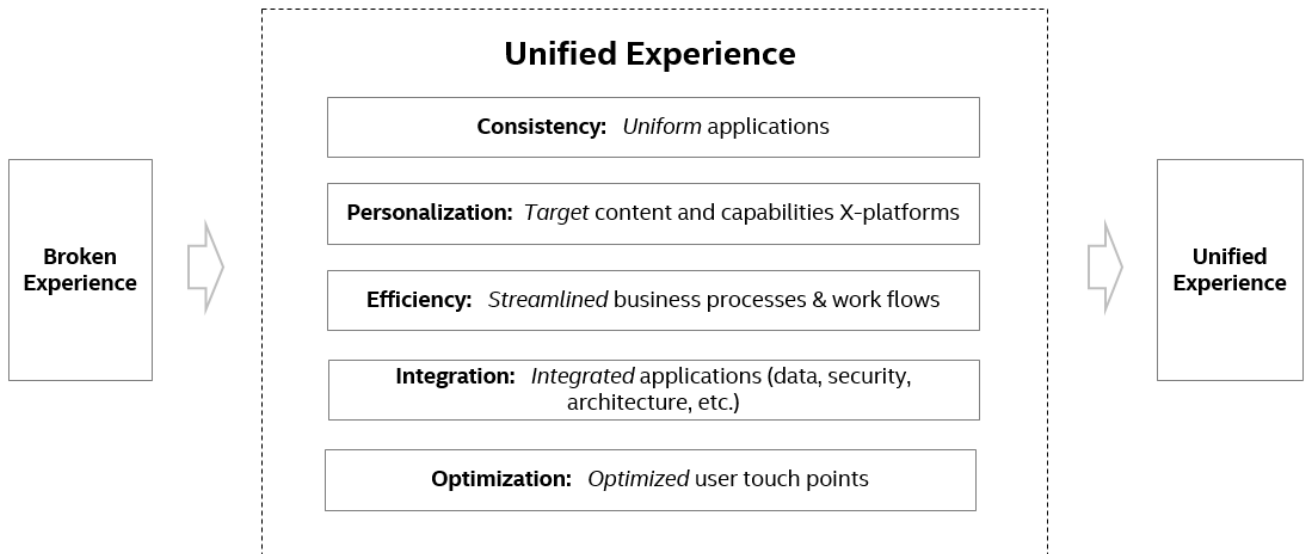


Figure 2. Definition of Unified Experience.

Our definition of unified experience has five attributes (see Fig. 2). These are consistency, personalization, efficiency, integration, and optimization. More specifically,

- 1) uniform applications help achieve *consistent* visual and branding design across applications;
- 2) streamlined business processes that ensure applications meet business goals with *efficiency* besides quality and cost savings.
- 3) target content and capabilities across platforms and devices deliver *personalized* content and capabilities for end users in the appropriate context as needed;
- 4) *integrated* applications guarantee the optimal integration design across applications in terms of data, security, architecture; and

- 5) *optimized* user touch points deliver a seamless end-to-end experience, including optimized design across the touch points that interact with applications, such as application performance (fast), capability (useful), usability (useful), marketing and communication (be aware and ready), user help (easy to get help), user support (easy to get support).

What a daily enterprise user, who interacts with a set of applications in their daily routine work, gets from a unified experience is:

- 1) a suite of applications with consistent visual/branding style and interactions on the UIs;
- 2) streamlined business processes and work flows behind applications;
- 3) personalized target content and capabilities across platforms and devices, applications know who they are, what platforms/devices they are using and automatically feed information as they go in the right context;
- 4) the ability to easily accomplish business tasks across applications because fewer applications are needed without any need for concern about interruption in data, security, etc.; and
- 5) they can easily set up/install applications when needed, come to a consolidated digital workplace to access the applications needed for their jobs, easily access support and help when needed since they benefit from optimized user touch points.

Norman and Nielsen's definition of UX is commonly accepted: "UX encompasses all aspects of the end-user's interaction with the company, its services, and its products" (Norman & Nielsen, 2015). A key point must be emphasized here in relation to our definition of unified experience. The unified experience here focuses on all the aspects of the end user's interaction with *multiple-applications* in a business context, where certain business processes, rules, logics, and dependencies are embedded across such a suite of applications. Users might have reasonably good experiences when using an individual application, but a unified experience won't be achieved if the design is not optimized across applications, which can happen if the interaction design across applications is not consistent, if the business processes across applications are not streamlined, or if the data quality across applications is poor due to lack of data integration across multiple databases.

Our definition of unified experience represents a systematic view of delivering end-to-end experience to target users in the solutions. Also, the definition of unified experience represents our vision—an ideal state that UX professionals and companies should pursue. Because of this, we recognize the challenges on the road towards this final state.

3.2 A Implementation Framework for Unified Experience

To help achieve unified experience, we propose an implementation framework (see Fig. 3). Table 1 describes the details of each element of the framework.

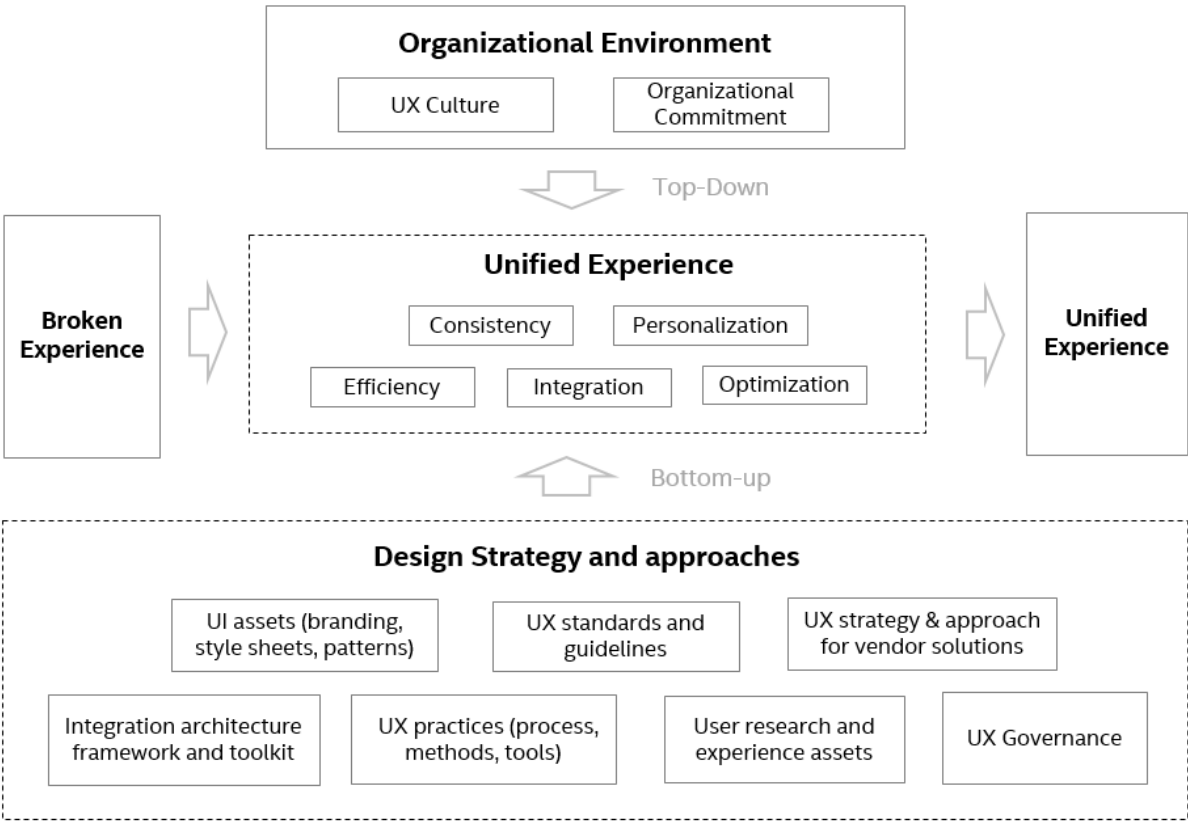


Figure 3. Implementation Framework for Unified Experience.

Table 1. Description of the Unified Experience Framework.

Components	Descriptions (Primary Examples of Assets & Activities)	Goals

Driving Vector 1 – Organization		
UX culture	Experience scorecard, UX maturity model.	Prioritize UX investment; drive actions based on maturity.
Organizational commitment	Streamline business processes as high priority, drive UX-based prioritization process of investment, and execute strategic procurement of vendor solutions.	Support activities for unified experience.
Driving Vector 2 - Design		
UX standards	UX design standards & guidelines, business process redesign guidelines.	Support UX design and business process design.
UI assets	Reusable UI assets library: style sheets, branding assets, interaction design patterns, code-based UI components.	Build uniform UI across apps.
Integration framework	Integration architecture, UX architecture, API (application programming interface) services, toolkits.	Guide integration across apps.
UX practices	Processes of Design for Unified Experience, methods, agile development, and tools.	Guide organization to achieve unified experience.
Strategy for vendor solutions	Maintain less number of major vendors through consolidation; UX assessment during vendor selection; strategic influences of vendor design through collaboration.	Optimize UX design in vendor solutions.
Governance	Governance process, checklists, automatic tools (e.g., code scanning tools).	Facilitate and guide practices.

As illustrated by Fig. 3, the implementation framework defines two primary driving vectors for achieving unified experience: organization and design. The organization driving vector includes UX culture and organizational commitment, which is basically a top-down approach that would create a mature organizational culture as a foundation to deliver unified experience (see detailed description in Table 1 for the framework). As a bottom-up approach, the design driving vector, as defined in the framework, provides the necessary processes, tools, and assets in order to enable and facilitate implementation.

The implementation framework encourages both strategic and tactical approaches. Strategically speaking, a top-down approach is preferred and has greater impact, but it requires a significant shift in organizational culture and a strong level of commitment, which, in many cases, may not be realistic (Klocek, 2012). The tactical approach begins with surface improvements in UI and

visual design, goes progressively deeper into structural items, and ends with a large-scale strategic organizational shift.

The framework encourages companies to take an adaptive and flexible approach. Based on the level of UX maturity within an organization, the implementation of unified experience could be executed with a phased approach, starting from either side (i.e. top-down or bottom-up) or from the middle layer of business process redesign or integration design for selected business domains. UX professionals within a company may take any route through the framework according to their needs and resources.

As part of our UX vision, we recognize the difficulty of achieving the defined unified experience by fully implementing the five attributes in a suite of applications (see Fig. 2). One may consider the five attributes as the variables in the unified experience formula. The amount each attribute contributes to the overall formula may vary, depending on factors such as complexity, business cost/priority, and end users' experience maturity in applications. For instance, if business process and integration design have the biggest impacts on business value and UX in selected domains, we may consider marking these two areas as high priority, whenever such a prioritization choice has to be made.

Specifically, in the hybrid environment of enterprise software where vendor solutions are heavily deployed, for instance, it is likely that in the short-term we would only achieve a modest level of unified experience across vendor solutions. We would not be able to achieve a perfectly uniform experience (e.g. consistent UI) because people cannot expect all enterprise software vendors to adopt exactly the same visual design and interaction models, or provide UI configuration capabilities for their corporate customers to customize the UI according to their own visual and branding requirements. However, we will do our best to ensure that the overall unified experience is improved through other means such as efficiency (streamlined workflows), integration, personalization, and optimized user touch points. A longer term strategy for UX would be for a company to adopt fewer vendor solutions (platforms). That way the UX landscape of enterprise software within a company would end up with a lower number of major vendor platform-based experiences, plus necessary home-grown solutions for specific business needs in order to maximize the contribution of consistency and integration to the overall unified experience.

3.3 Comparison between our Approach and Previous Approaches

Compared to approaches proposed by others, our definition of unified experience and the implementation framework are characterized by the following features.

- **An end-to-end systematic view.** Our approach covers the whole spectrum (the five attributes) of end-user interaction with applications, including front-end, business process, backend, integration, personalization, platform, and devices. The goal is to achieve an end-to-end unified experience.
- **Holistic.** We promote UX work that addresses all user touch points that interact with applications, including set-up/installation, user support, user help, communication and marketing, so that applications are easy to set up/install, easy to learn, and easy to get support.
- **More realistic and specific.** To face reality, we provide specific strategic and tactical approaches for achieving unified experience in a hybrid environment. Specifically, we design strategies for working on vendor solutions. Although we focus on the enterprise software domain, the approach may be extended beyond it.
- **More adaptive and opportunistic.** We define two streams in the framework (top-down and bottom-up). People could execute the framework from either direction based on the maturity level of a company, or start from any layer within the framework using a phased approach.
- **UX metric driven.** We recognize the business value (return of investment) of UX work. We recommend a UX metric-driven approach. Business and technology partners must see benefits before taking a proactive approach. The metric-driven approach also allows management to prioritize investment based on the UX scorecard, such as user productivity, support cost, and user satisfaction.
- **More tactical.** The framework calls for specific approaches and tools to achieve unified experience. For instance, determining the UX design approach for vendor solutions (e.g. partnership with vendors and vendor UX assessment) and building UX assets for home grown applications (e.g. reusable UI assets).

4. CASE STUDY

4.1 Problem Statement

Let's meet Paul who is a design persona in this case study. Paul is a buyer who works for a global company's supply chain organization. He has been working there for 10 years, with 3 and a half years' experience in his current role. He holds a bachelor degree and manages an inventory of factory spare parts. His day is mostly driven by tactics and exceptions. He is struggling to prioritize and execute tasks in the right sequence, which leads to major operational issues. For example, a machine might be down awaiting a spare part, in which case senior management escalation and expedited shipping might incur huge cost. Various systematic tools generate multiple signals about upcoming risks, but significant manual effort is necessary to filter out the noise and discover what the real problem is. Paul is required to take the appropriate actions to put out the fire. The amount of data is overwhelming and there are numerous transactional systems that Paul must interact with in order to resolve the issue. The UX is poor and Paul is expected to have tribal knowledge of the processes and systems that keep the business running.

Paul is the liaison between the factory and the supplier. He passes decisions and information between them, which can occasionally cause delays. Paul and his team are spending thousands of hours on gathering and analyzing data from multiple sources and executing decisions in Enterprise Resource Planning (ERP) and Material Requirements Planning (MRP) systems. All in all, Paul is very frustrated with these processes and the tools that are supposed to support his job. He wishes to have "one comprehensive system for all transactions instead of dozen different systems." Ideally, if Paul had more time, he would spend it on strategic rather than tactical matters. This would include proactive supplier management and cost saving projects.

Several actions have been undertaken in order to solve Paul's problem. A cause and effect analysis shows that there are three major issues. These problems are summarized in Table 2.

Table 2: Summary of Main Buyer Problems

Issue	Description	Impact	Severity
Long Cycle Time	Long Waiting time + long processing time	Productivity – requires high headcount, late orders, UX	High
Incorrect Priorities	Critical items not being worked as top priority	Factory escalations, UX	High
Quality	Transactions are not processed correctly	Rework, cost, UX	High

4.2 Approach

Our approach was to follow the implementation framework for unified experience described above (see Figure 3). The objective was to create an end-to-end experience for the buyer by providing comprehensive system capability that considerably simplified the UI while seamlessly supporting the business process. According to the analysis, it was clear that the buyer required a one-stop shop for carrying out his job effectively. The vision was to design a system that recommends actions, streamlines decision-making, and increases productivity.

The focus was on the three attributes that comprise unified experience as defined in Figure 2: *business process redesign*, system-to-system *Integration*, and user touch point *Optimization*. The scope of work included business process reengineering and digitization, resetting roles and responsibilities, standardizing process measures and metrics, maximizing process output, reducing manual work, and enabling self-service for customers of the process.

4.2.1 Initial Analysis

The initial analysis was conducted across the three attributes of unified experience: business process redesign, integration and optimization. First, when we examined the buyer's business process we realized that they were constantly fighting fires in order to sustain the operations without success. We noticed that the dynamic nature of the environment made it difficult for the buyer to keep track of everything. The buyer was expected to shift priorities from one task to another, even when the first remained incomplete or imperfectly executed. This resulted in a large backlog of work, inefficient prioritization, and a long waiting time that on occasions escalated to senior management intervention.

We also found that the buyer was jumping between various applications in order to complete his tasks successfully. Some of this was merely copying and pasting data between different screens, but it could also mean that the data had to be entered or analysis had to be carried out on another application. This frequently saw the buyer lose focus because the original matter was difficult to memorize. Moreover, the buyer was forced to perform numerous mouse clicks, introducing ergonomic problems. Further, there was a constant pressure from the factory stakeholders who wished to be serviced as quickly as possible. The result was an increase in the time taken to complete the task, issues with quality, tasks having to be reworked, a loss of productivity, and frustration.

Secondly, from an integration perspective, the buyer had to bring together all the data elements from the various systems in order to support decision making. In order to understand the problem better, we mapped out all the systems from which the buyer was retrieving data. Additionally, we listed any transactional system that the buyer used to perform actions after making a decision, such as creating a Purchase Order (PO). The objective was to integrate the disparate modules into a single system through which the entire business process flowed; this would display all the information required in one place, allowing decisions to be made via a simple button click.

Lastly, we carried out an initial analysis on user touch points. We realized that the buyer had to repeat a number of tasks. For example, the buyer had to send an email to the supplier asking for a status update on a certain PO multiple times per day. This required the buyer to populate a pre-defined email template with the specific PO information. Monitoring the supplier response fell outside the system, but still mandated system action. Due to the complexity of the task, the buyer made occasional mistakes.

4.2.2. Design Principles

Designing a unified experience not only required a paradigm shift, but also a transformation of the operation from reactive handling to proactive management of inventory. Several design principles were developed before we designed the new “Bill of Materials Application” (BOMA), which were applied as the design was carried out (see Table 3).

Table 3: Summary of Design Principles

#	Principle	Description	Benefits
1	Standardize the process where possible	Create a single version of the process while accepting the variants when standardization is not possible	UX, productivity or any other process measures (e.g. cost saving, risk avoidance, etc.)
2	Integrate Business Intelligence (BI)	Display the minimum information required for making a decision	Productivity, UX, decision making
3	Reduce the number of siloed systems	Integrate information from external systems to avoid switching context too frequently	Concentration on task, accuracy
4	Collaborate online	Digitize the interaction between roles	Faster response times, avoiding rejection
5	Eliminate rework	Do it right the first time	Productivity, quality
6	Allow customers to serve themselves	Expose the state of current process customers	Builds trust, reduces manual non-value add work

Below we provide an explanation for each design principle.

- *Principle 1 – standardize the process where possible:* Standardized business processes contribute to process effectiveness (speed), quality (minimizing defects per unit produced), and flexibility (the ability to make changes). On the other hand, standardization was not possible in every case. We accepted the process deviations when it was impossible to standardize them or it would have over-complicated the process.
- *Principle 2 – Integrate Business Intelligence (BI):* We brought the minimum BI required to the decision point, a single screen. This meant applying data transformations and business rules, as well as summarizing the conclusion in a usable, readable, simple format on the screen so that any user (new or advanced) could make decisions quickly without investing a lot of time in data analysis.
- *Principle 3 – Reduce the number of siloed systems:* We reduced the number of applications the user interacts with by integrating data from multiple data sources.
- *Principle 4 – Collaborate online:* In order to optimize user touch points, we empowered the user with a social platform that enabled them to collaborate directly instead of having to use emails. The data entered was used to drive further automation.

- *Principle 5 – Eliminate rework:* We designed the process so that any data is systematically verified at the point of entry, preventing rejections at a later point.
- *Principle 6 – Allow customers to serve themselves:* We improved the service that buyers provide to their stakeholders by enabling a self-service report that reflects the buyer's work. This avoids non-value added phone-call or emails and helps to optimize user touch points.

4.2.3. Designing a To-Be Process

Redesigning the business process was a crucial part of our work. Initially, we mapped out all the process activities and classified them as manual or systematic. The manual steps then became candidates for automation. Secondly, we identified all the roles involved and evaluated which were redundant and could safely be removed. Finally, we looked at the decisions made at each step and the logic utilized to make decisions. The information was used for automation.

We used a combination of the following techniques:

1. *Direct Observation* – We attempted to understand what the users were doing by watching them doing their job. We asked some of the buyers to walk us through the flow (active); in other cases we sat behind them without disturbing them and took notes (passive).
2. *Data Collection* – To get a broad understanding of the problem statement we conducted a survey. We also ran plenty of reports to gain more insights on the process behavior, the effectiveness of the role and the amount of rework they generated.
3. *Data Validation* – We verified our to-be process proposals with process experts and we ran simulations on live transactional data to validate them.

The results of this work generated a list of user pain points, a more accurate problem statement, and a lot of data on the process (e.g. flow, role, logic). Based on the data, we discovered what could be automated, what roles could be removed, what data and systems needed to be integrated, what activities could be excluded, and how the users could become proactive.

We worked closely with the process stakeholders on the to-be design. We came up with countless proposals about how to tackle each of the problems. The whole redesign process was iterative. We started with a simple version and revised it continuously until we couldn't optimize it further. Fig. 4 shows an example of the to-be process for BOMA.

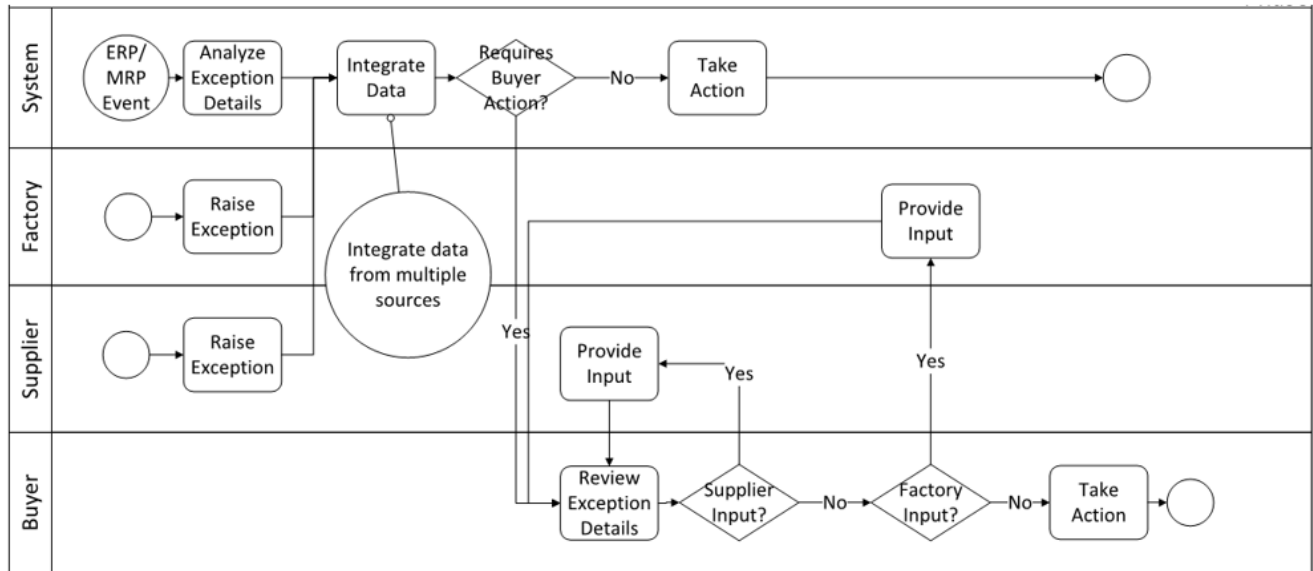


Figure 4: An Example of the To-Be Process for BOMA

4.2.4. Implementation

The technical enabler for the to-be redesign was a Business Process Management Suite (BPMS). BPMS is a software development product that offers business process management capabilities—business process modeling and orchestration, a business rules engine, collaboration, integration and Business Activity Monitoring (BAM), to name a few.

Implementing the to-be process by incrementally improving the current systems landscape was impossible due to the uniqueness of the unified experience process needs:

- Integration – at the heart of the process we had a strict ERP system that had to interact with other systems. The ERP couldn't be customized and adapted to enable such integration because the cost of maintenance would have increased substantially.
- Business Process Management (BPM) Capabilities – the to-be process relied heavily on process management capabilities. The standard enterprise solution stack did not offer such capabilities out of the box.

The BPMS workflow integrated data from a variety of sources through a Service Oriented Architecture (SOA) layer. This layer allowed services to be abstracted or the exchange of data

between the workflow and other systems. The greatest benefit of this architecture was the flexibility for changes when compared to the strict ERP or MRP.

The main flow of the BPMS tool we developed was triggered by a system (e.g. ERP) or a human event (e.g. a factory user), integrated data and performed calculations. Eventually, the user would be able to open the result in a user interface where they could review the system's recommendations, analyze the contextual data, and be able to make a decision that integrated data back into the source system.

The new tool provided exclusive process Key Performance Indicators (KPI's). In the as-is architecture the timing of the process tasks was not captured systematically, since a lot of work is done outside the system. The to-be architecture exposed how the process is behaving (activity metrics, team's performance, etc.) and drove continuous initiatives to improve the process and the design.

4.3 Results and Benefits

Table 4 summarizes the results of the improved UX across business processes, integration, and optimization. It shows that the process redesign led to a comprehensive reduction of manual steps, eliminated an integrator role responsible for manual data analysis, decreased the rework rate, and consolidated seven manual processes into one. The integration resulted in much successful automation, and reduced the number of applications and screens the user had to interact with. Lastly, optimization enabled the user to interact with the system effectively as well as learn how to use it rapidly.

Table 4: Summary of Improved UX

Category	Indicator	Before	After
Process redesign	Number of Manual Steps	35	12
	Manual Processes	7	1
	Number of Roles	5	4
	Percentage of Rework	20%	<5%
	Number Proactive Processes	0	1
Integration	Number of Applications	6	1
	Number of Screens	15	3
	Number of Automated Steps	8	23
Optimization	Manual Emails	Dozens	None
	Online Help	Partial, inconsistent	Single, comprehensive
	Respond to Customers' Queries	Slow response	Fast

Three months after the new system was deployed, we collected data to assess the differences (before and after). As shown in Table 5, the unified experience framework enabled a number of business value benefits to be achieved, such as responding faster to risk, for example. Since the new process encompasses the entire flow, it's easier to make decisions quickly and to substantially decrease risk. Not surprisingly, the users appreciate the new capabilities, which are articulated in many other measures, such as the average number clicks per transaction, the ease of use, the time take to reach competence, and the system's learnability.

Table 5. Benefits of Unified Experience Design

Indicator	As Measured by	Before	After
Productivity	Average time taken to complete a task	10 minutes	5 minutes
Risk indicator	Number of items that may not be available once required divided by the total number of items	0.85%	Less than 0.66%
Late orders	Number of orders that do not arrive on time because of a supplier delay divided by total number of orders	40%	20%
Cost savings	US dollars each buyer is saving based on a proactive planning of inventory	\$0	\$1.2 Million
Ergonomics	Average number of clicks per transaction	69	17
Ease of use	Feedback of the users and process experts	Frustration	Enthusiastic
Time to competence	Hours spent in initial training, when learning to work with the new system	Dozens hours	2 hours
System learnability	Volume of "How do I" tickets	Dozens	0
Next best action	% of decisions made agreeing with system's recommendation	<10%	83%

5.1 DISCUSSIONS AND CLOSING COMMENTS

The ultimate goal of unified experience is to deliver business value. In the end, UX professionals have to prove that these changes achieve improved revenue, efficiency, employee productivity, effective decision-making, employee satisfaction and retention, and any other measure that can be used to justify doing the work and "prove" that the change is due to UX.

To promote the approach proposed here, UX professionals have to measure the value of unifying the UX of products. Demonstrating the benefits of UX work has been a challenge for UX professionals. Tactically speaking, they should choose the parameters that can be easily measured and are thus more likely to be accepted by business and management as justifying UX work. As the case study demonstrates, there are quantitative metrics that allow us to readily demonstrate the improvements UX work can bring (see Table 5). The work shared in this case study exemplifies how the approach delivers high business value and fantastic improvements in UX.

Strategically speaking, a critical component of our metric-driven approach (see Fig. 3) is an experience scorecard at a corporate level, which helps to measure the return on the investment made in UX. The experience scorecard can also help upper management to monitor and identify the gaps in unified experience across various business units and portfolios within a company, which could guide the prioritization of investment in unified experience. Sample UX metrics include support call volume and root causes, app real-time UX analytics, and user surveys.

5.2 Implementation Strategy

As discussed earlier, there is no a fixed way to implement unified experience. Although we expect a top-down approach so that we can leverage a mature organizational culture and commitment to drive the work, in reality we have to employ a phased approach to implement the approach in selected domains.

Within the Intel IT organization, we have made progress towards a unified experience. Over time, efforts have been taken to address unified experience by tackling individual attributes. In order to deliver a uniform experience across applications, we have built reusable UI assets, such as style sheets built with Intel branding identities and UI components. These UI assets have been shared and reused across many home-grown applications and vendor applications whenever a vendor's UI configuration has been suitable. Also, several code-based UI components using Angular JavaScript were developed for internal reuse. These UI components have helped achieve consistent interactions across many home-grown applications.

In order to address the personalization attribute of unified experience, we have published UX design guidelines for web responsive design (WRD) and mobile app design, including a UI

design pattern library for WRD and mobile UI interaction patterns (Appel, & Xu, 2014; Xu, 2014B). The intention is to share and reuse effective interaction patterns across desktop and mobile applications so that users receive target content and capabilities in a usable UI format across various screen sizes and devices.

In addition, a user experience design (UXD) approach has been practiced in projects that have been carried out within the IT organization (Xu, 2014A). UX professionals do not just focus on UI design and its usability, but also optimize all the user touch points, including marketing and communication, user help, user support, and change management.

The case study presented above focuses on business process, integration redesign, and optimization of user touch points. As people are moving forward from the surface improvement (e.g., UI layer) to the deeper areas such as business process and integration redesign, there will be more challenges and but more return of investment if the redesign is done correctly. This has been proved by previous studies (Davenport & Short, 1990; Steen & Schmidt, 2007; O’riain, et al., 2012). Kloczek (2012) recognizes the importance and challenges when designing a unified experience: “It could lead to collapsing multiple applications into one, bridging gaps with new applications, eliminating redundancies in capabilities or refocusing the service. This kind of work takes deep organizational commitment and a strong mandate.”

The case study also demonstrates a phased approach. Given the BOMA project success, organizations should evaluate how to deploy a similar approach on a broader scope by consolidating redundant organizational functions and processes. In addition, selecting the right technology to enable redesigned business processes can help to expedite the delivery of the improved solution that involves business process and integration redesign. In our case study, the BPMS product is a core component of the architecture that enabled the implementation of the new solution.

In order to implement unified experience, UX professionals must lead the initiatives by closely partnering with business and technical people (Rogowski, 2012). A UX organization within a company must take the lead in working within and across silos to define a clear unified experience strategy and building repeatable design processes. Organizations need to develop their disciplines and governance processes into repeatable practices that help tie experiences together. Also, UX professionals need to document and disseminate information about the end

user's singular experience by building tools such as central repositories of personas and user-journey maps.

UX professionals within companies need to facilitate cross-silo coordination (Abel, 2015). Although fixing the silos is a long-term objective, it won't happen without major organizational changes. In the meantime, UX professionals need to serve as educators for siloed teams that typically don't communicate with each other. A maturity model of unified experience can be developed to assess the maturity level of business units, in combination with an experience scorecard. By doing so, UX professionals will have more "weapons" with which they can reinforce the necessary change in organizational culture and organizational commitment.

In summary, this chapter presents a new perspective on unified experience with an implementation framework and a case study. Although we discuss the challenges and recommend an approach within the context of enterprise software, unified experience should be the ultimate goal, not just for the enterprise software, but also for the broader external consumer world.

ACKNOWLEDGEMENTS

We would like to thank Jennifer Coyne, Jerzy Bilchuk, Bala Suresh, and Hardev Chouhan for their input for developing the unified experience framework. The case study work described in this chapter is a cumulative experience of several people; we would like to thank Stephen Healy, Anand Kumar, Yaoz Talshir, Robert Muller, Ekaterina Potapova, and Aakash Gaur for their continued contributions developing the approach. Any opinion, findings, conclusions, or recommendations expressed herein are those of the authors and do not necessarily reflect the views of other individuals or any corporations

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